

Show understanding of how data for a bitmapped image are encoded

9618_w25_qp_11_sc_7.d.ii

QUESTION

(ii) Explain why the actual file size may be larger than the one calculated in (d)(i).

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..... [2]

MARK SCHEME

7(d)(ii)	1 mark per bullet point, max 2 marks	2
	• The file will contain metadata • The file will have a header	

QUESTION

3 A computer stores images and text files.

(a) One of the images is a bitmapped image.

Complete the table by writing the answer for each statement.

Statement	Answer
the term for the smallest element that makes up an image	
the largest number of different colours that can be represented with a bit depth of 8 bits	
the term for the dots per inch (dpi) when an image is displayed	

[3]

MARK SCHEME

3(a)	1 mark for each correct answer	3								
<table><tr><th>Statement</th><th>Answer</th></tr><tr><td>The term for the smallest element that makes up an image.</td><td>pixel</td></tr><tr><td>The largest number of different colours that can be represented with a bit depth of 8 bits.</td><td>256 // 2⁸</td></tr><tr><td>The term for the dots per inch (dpi) when an image is displayed.</td><td>Screen resolution</td></tr></table>		Statement	Answer	The term for the smallest element that makes up an image.	pixel	The largest number of different colours that can be represented with a bit depth of 8 bits.	256 // 2⁸	The term for the dots per inch (dpi) when an image is displayed.	Screen resolution	
Statement	Answer									
The term for the smallest element that makes up an image.	pixel									
The largest number of different colours that can be represented with a bit depth of 8 bits.	256 // 2⁸									
The term for the dots per inch (dpi) when an image is displayed.	Screen resolution									

QUESTION

(b) The colour of each pixel in a bitmapped image is represented by 8 bits.

(i) State the largest number of different colours that can be represented by 8 bits.

..... [1]

MARK SCHEME

6(b)(i)	1 mark for: 256 // 2^8	1
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9618_w24_qp_12_sc_7.a.i

QUESTION

7 A student takes a photograph of a science experiment.

(a) The photograph is saved as a bitmapped image.

(i) Define the following bitmap terms.

Colour depth

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File header

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[2]

MARK SCHEME

7(a)(i)	1 mark for each correct definition Colour depth: - the number of bits used to represent a colour // the number of colours that can be represented in an image File header: - stores data about the image file / metadata	2
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QUESTION

(d) The computer can transmit a video made from bitmap images and vector graphic animations to the VR headset.

(i) Describe how the data for a bitmapped image is encoded.

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..... [3]

MARK SCHEME

2(d)(i)	1 mark each: - Image is made of pixels and each pixel has one colour - Each colour has a unique binary code Code for the colour of each pixel is stored in sequence	3
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QUESTION

(b) State what is meant by the **bit depth** of a bitmap image **and** explain how changing the bit depth affects the image.

Definition

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Explanation

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..... [3]

MARK SCHEME

1(b)	1 mark for the definition - The number of bits used to represent each colour 1 mark for each bullet point for the explanation - Increase in bit depth means the image has a greater range of colours // Decrease in bit depth means the image has a smaller range of colours - Increase in bit depth makes the image closer to the original / more realistic // Decrease in bit depth makes the image less like the original / less realistic	3
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9618_s23_qp_11_sc_1.a

QUESTION

- 1 Images are being created to advertise holidays.

Some of the images are bitmap images and some are vector graphics.

- (a) Complete the table by defining the image terms.

Term	Definition
Drawing list	
Pixel	
Colour depth	

[3]

MARK SCHEME

1(a)	1 mark for each correct definition	3								
	<table><tr><th>Term</th><th>Definition</th></tr><tr><td>Drawing list</td><td>All the drawing objects in an image // a list that stores the commands required to draw each object</td></tr><tr><td>Pixel</td><td>The smallest part of the image // one square / dot of one colour</td></tr><tr><td>Colour depth</td><td>The number of bits per pixel // determines the number of colours that can be represented in the image</td></tr></table>	Term	Definition	Drawing list	All the drawing objects in an image // a list that stores the commands required to draw each object	Pixel	The smallest part of the image // one square / dot of one colour	Colour depth	The number of bits per pixel // determines the number of colours that can be represented in the image	
Term	Definition									
Drawing list	All the drawing objects in an image // a list that stores the commands required to draw each object									
Pixel	The smallest part of the image // one square / dot of one colour									
Colour depth	The number of bits per pixel // determines the number of colours that can be represented in the image									

9618_s23_qp_11_sc_1.b.i

QUESTION

(b) The bitmap images are photographs of the holiday locations.

(i) Colour depth and image resolution are both included in the file header of a bitmap image.

Identify **two other** items that could be included in the file header of each photograph.

1

2 [2]

MARK SCHEME

1(b)(i)	1 mark each Examples: • Confirmation that it is a bitmap // file type • Compression type • Location/offset of data within the file • Dimensions e.g. 100 × 100 pixels	2
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QUESTION

(e) The photographs are stored as bitmap images.

Complete the statements about bitmap images by writing the missing words.

The of a bitmap image is the number of bits that are used to store each pixel.

Metadata about the image is stored in the of the file.

[2]

MARK SCHEME

2(e)	1 mark for each correctly completed term; The bit depth of a bitmap image is the number of bits that are used to store each pixel. Metadata about the image is stored in the header of the file.	2
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QUESTION

(b) Riya takes a photograph using a digital camera. The photograph is stored as a bitmap image.

(i) Describe **two** differences between a vector graphic and a bitmap image.

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2

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[4]

MARK SCHEME

5(b)(i)	1 mark for each bullet point to max 2 for each difference • Bitmap made up of pixels // bitmap is made of colours stored for individual pixels • Vector graphic store a set of instructions about how to draw the shape • When bitmap is enlarged the pixels get bigger and it pixelates • When vector is enlarged it is recalculated and does not pixelate • Bitmap files are usually bigger than vector graphics files because of the need to store data about each pixel • Vector graphics have smaller file size because they contain just the instructions to create the shapes • Bitmap images can be compressed with significant reduction in file size • Vector graphic images do not compress well because of little redundant data	4
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9618_s21_qp_11_sc_1.a.i

QUESTION

1 Anya scans an image into her computer for a school project.

(a) The scanned image is a bitmapped image.

(i) Complete the following table to describe the two terms about graphics.

Term	Description
Pixel	
File header	

[2]

MARK SCHEME

1(a)(i)	1 mark for each description Pixel: • A single square of one colour • The smallest addressable element in an image File header: • Data about the bitmap image (e.g. number of colours)	2
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9618_w25_qp_13_sc_7.b

QUESTION

- (d) A bitmap image has a resolution of 1000 pixels wide by 2000 pixels high. The colour depth is 16 bits.

Calculate an estimate of the file size in megabytes.

Show your working.

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File size megabytes

[2]

MARK SCHEME

6(d)	1 mark for the working 1 mark for the correct answer Working: $(1000 * 2000 * 16) / (8 * 1000 * 1000)$ $// (1000 * 2000 * 2 / (1000 * 1000))$ Answer: 4 megabytes	2
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QUESTION

(d) A bitmap file is downloaded. The image has a maximum of 256 colours and measures 512 pixels wide by 2048 pixels high.

(i) Calculate an estimate of the file size of the bitmap in kibibytes.

Show your working.

Working

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File size of bitmap kibibytes

[2]

MARK SCHEME

7(d)(i)	1 mark for the working 1 mark for the correct answer Working: $(512 * 2048 * 8) / (8 * 1024) // (512 * 2048) / 1024$ $// (2^9 * 2^{11} * 2^3) / (2^3 * 2^{10}) // (2^9 * 2^{11}) / 2^{10}$ Answer: 1024 kibibytes // 2^{10} kibibytes	2
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QUESTION

(ii) Explain why the actual file size may be larger than the one calculated in (d)(i).

.....

.....

.....

..... [2]

MARK SCHEME

7(d)(ii)	1 mark per bullet point, max 2 marks - The file will contain metadata - The file will have a header	2
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QUESTION

- 1 (a) A student paints a picture in an art class.

The student takes a photograph of the picture using a digital camera. The digital camera creates an image with a resolution of 2 million pixels and uses a bit depth of 16 bits.

- (i) Calculate the file size of the image created by the digital camera in megabytes (MB).

Show your working.

Working space

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Answer MB

[2]

MARK SCHEME

1(a)(i)	1 mark for correct working: e.g. $2000000 \times 16 / (8 \times 1000 \times 1000)$ 1 mark for answer: 4MB	2
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QUESTION

3

- 2 A photograph is stored as a bitmap image.

- (a) The photograph has a resolution of 4000 pixels wide by 3000 pixels high. The bit depth is 4 bytes.

Calculate an estimate for the file size of the photograph in megabytes.

Show your working.

Working

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Answer megabytes

[2]

MARK SCHEME

2(a)	1 mark for working: • $4000 * 3000 * 4$ 1 mark for correct answer: • 48MB	2
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9618_w23_qp_13_sc_2.b

QUESTION

(b) Calculate the file size of a bitmap image using the following information:

- image resolution of 2048 pixels wide and 1024 pixels high
- bit depth of 16 bits.

Give your answer in kibibytes. Show your working.

Working

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Answer in kibibytes [2]

MARK SCHEME

2(b)	1 mark for answer. 1 mark for working Answer: 4096 kibibytes Working: File size = $(2048 \times 1024 \times 16) / (8 \times 1024)$	2
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QUESTION

(c) A bitmap image has a resolution of 2048 pixels wide and 1024 pixels high.

The image has a bit depth of 10 bits per pixel.

Estimate the file size of the bitmap image in mebibytes. Show your working.

Working

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Estimated file size in mebibytes

[2]

MARK SCHEME

6(c)	1 mark for answer, 1 mark for working Answer: 2.5 mebibytes Working: $(2048 \times 1024 \times 10) / (8 \times 1024 \times 1024)$	2
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QUESTION

(ii) One of the photographs has a bit depth of 8 bytes and an image resolution of 1500 pixels wide and 3000 pixels high.

Calculate the file size of the photograph in megabytes. Show your working.

Working

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File size MB

[2]

MARK SCHEME

1(b)(ii)	1 mark for working; 1 mark for answer - Working: $(1500 * 3000 * 8) / 1000 / 1000$ - Answer: 36 MB	2
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9618_s21_qp_11_sc_1.a.ii

QUESTION

(ii) The image is scanned with an image resolution of 1024×512 pixels, and a colour depth of 8 bits per pixel.

Calculate an estimate for the file size, giving your answer in mebibytes. Show your working.

Working

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Answer mebibytes

[3]

MARK SCHEME

1(a)(ii)	1 mark per bullet point for working, 1 mark for answer Working: $1024 \times 512 = 524\,288$ pixels/bytes $524288 / 1024 / 1024$ Answer: 0.50 mebibytes	3
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Show understanding of the effects of changing elements of a bitmap image on the image quality and file size

9618_s25_qp_13_sc_1.a.ii

QUESTION

(ii) The student takes a second photograph with a lower bit depth.
Explain the effect of decreasing the bit depth on the image and on the image file.
Effect on image
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Effect on image file
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MARK SCHEME

1(a)(ii)	1 mark each Image: - There will be fewer shades of colour available - so the image does not match the original as detail is lost Image file: - Fewer bits are used to store each pixel - so less data is stored, therefore the file size is reduced	4
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9618_w24_qp_13_sc_6.b.ii

QUESTION

- (ii) State **one** drawback of increasing the number of bits that represents each pixel in the bitmap image.

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..... [1]

MARK SCHEME

6(b)(ii)	1 mark for: Increased file size	1
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9618_w24_qp_12_sc_7.a.ii

QUESTION

- (ii) Explain why changing the image resolution will affect the image quality and file size.

Image quality
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.....
File size
.....
..... [2]

MARK SCHEME

7(a)(ii)	1 mark for each correct explanation Image quality: - Decreasing resolution means details within the image are lost because there are fewer pixels // Increasing resolution means the image is more detailed because there are more pixels File size: - Decreasing the resolution will decrease the file size because there are fewer pixels therefore less data // Increasing the resolution will increase the file size because there are more pixels therefore more data	2
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QUESTION

(c) Identify **two** elements of a bitmap image that can be changed to reduce its file size.

- 1
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- 2
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- [2]

MARK SCHEME

2(c)	1 mark each to max 2 : • Colour/bit depth • Image resolution	2
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QUESTION

2 (a) Describe the impact of increasing the image resolution on the quality of a bitmap graphic.

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- [2]

MARK SCHEME

2(a)	1 mark for each bullet point: • more pixels can be stored / are available • the image is sharper / less pixelated	2
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QUESTION

- (b) State what is meant by the **bit depth** of a bitmap image **and** explain how changing the bit depth affects the image.

Definition

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Explanation

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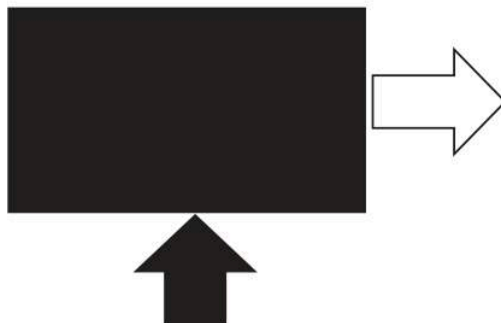
[3]

MARK SCHEME

1(b)	1 mark for the definition - The number of bits used to represent each colour 1 mark for each bullet point for the explanation - Increase in bit depth means the image has a greater range of colours // Decrease in bit depth means the image has a smaller range of colours - Increase in bit depth makes the image closer to the original / more realistic // Decrease in bit depth makes the image less like the original / less realistic	3
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QUESTION

- 8 The following bitmap image has a resolution of 4096×4096 pixels and a colour depth of 24 bits per pixel.



The image is displayed on a monitor that has a screen resolution of 1920×1080 pixels.

- (a) Tick (✓) **one** box in each row to identify the effect of each action on the image file size.

Action	Increases the file size	Decreases the file size	No change to the file size
Change the colour depth of the image file to 16 bits per pixel.			
Change the screen resolution to 1366×768 pixels.			
Change the colour of the rectangle from black to red.			

[2]

MARK SCHEME

8(a)	1 mark one or two correct row(s). 2 marks for all three correct rows.	2																
	<table><tr><th>Action</th><th>Increases the file size</th><th>Decreases the file size</th><th>No change to the file size</th></tr><tr><td>Change the colour depth of the image file to 16 bits per pixel.</td><td></td><td>✓</td><td></td></tr><tr><td>Change the screen resolution to 1366 × 768 pixels.</td><td></td><td></td><td>✓</td></tr><tr><td>Change the colour of the rectangle from black to red.</td><td></td><td></td><td>✓</td></tr></table>	Action	Increases the file size	Decreases the file size	No change to the file size	Change the colour depth of the image file to 16 bits per pixel.		✓		Change the screen resolution to 1366 × 768 pixels.			✓	Change the colour of the rectangle from black to red.			✓	
Action	Increases the file size	Decreases the file size	No change to the file size															
Change the colour depth of the image file to 16 bits per pixel.		✓																
Change the screen resolution to 1366 × 768 pixels.			✓															
Change the colour of the rectangle from black to red.			✓															

Show understanding of how data for a vector graphic are encoded

9618_w24_qp_12_sc_7.b

QUESTION

(b)

The student draws a picture on paper that is scanned into the computer and saved as a vector graphic.

Define the vector graphic terms property and drawing list.

Property

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Drawing list

.....

[2]

MARK SCHEME

7(b)	1 mark for each correct definition Property: - an attribute of a drawing object // data about a shape // defines one aspect of the appearance of a drawing object Drawing list: - all the drawing objects/shapes in an image // stores the commands/descriptions / mathematical equations required to draw each object	2
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9618_s24_qp_12_sc_2.d.ii

QUESTION

(ii)

Describe the contents of a vector graphic drawing list.

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[2]

MARK SCHEME

2(d)(ii)	1 mark each to max 2 : - List of objects in the drawing - A list that stores the command/description/equation required to draw each object - Properties of each object e.g. the fill colour, line weight/colour	2
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9618_w23_qp_11_sc_1.a

QUESTION

1 (a) Draw **one** line from each vector graphic term to its most appropriate description.

Term	Description
drawing list	a component created using a formula
drawing object	defines one characteristic of a component
property	data required to create all components in the graphic

[2]

MARK SCHEME

1(a)	2 marks for all 3 lines correct 1 mark for 1 line correct	2
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Term	Description
drawing list	a component created using a formula
drawing object	defines one characteristic of a component
property	data required to create all components in the graphic

QUESTION

- 1 Images are being created to advertise holidays.

Some of the images are bitmap images and some are vector graphics.

- (a) Complete the table by defining the image terms.

Term	Definition
Drawing list	
Pixel	
Colour depth	

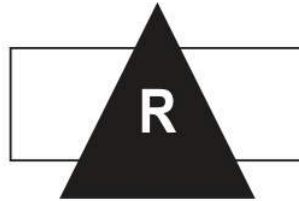
[3]

MARK SCHEME

1(a)	1 mark for each correct definition	3								
	<table><tr><th>Term</th><th>Definition</th></tr><tr><td>Drawing list</td><td>All the drawing objects in an image // a list that stores the commands required to draw each object</td></tr><tr><td>Pixel</td><td>The smallest part of the image // one square / dot of one colour</td></tr><tr><td>Colour depth</td><td>The number of bits per pixel // determines the number of colours that can be represented in the image</td></tr></table>	Term	Definition	Drawing list	All the drawing objects in an image // a list that stores the commands required to draw each object	Pixel	The smallest part of the image // one square / dot of one colour	Colour depth	The number of bits per pixel // determines the number of colours that can be represented in the image	
Term	Definition									
Drawing list	All the drawing objects in an image // a list that stores the commands required to draw each object									
Pixel	The smallest part of the image // one square / dot of one colour									
Colour depth	The number of bits per pixel // determines the number of colours that can be represented in the image									

QUESTION

- 5 Riya has created the following logo as a vector graphic.



- (a) Complete the table by writing a description of each vector graphic term **and** give an example for this logo.

Term	Description	Example from logo
Property		
Drawing list		

[4]

MARK SCHEME

5(a)	1 mark for each description, 1 mark for each valid example	4									
	<table> <tr> <th>Term</th><th>Description</th><th>Example from logo</th></tr> <tr> <td>Property</td><td>data about the shapes // defines one aspect of the appearance of the drawing object</td><td>e.g. black line // white fill // black fill //solid (line) // font of letter // colour of triangle</td></tr> <tr> <td>Drawing list</td><td>the list of shapes involved in an image // a list that stores the command/description required to draw each object</td><td>e.g. triangle // capital letter R // rectangle // line</td></tr> </table>	Term	Description	Example from logo	Property	data about the shapes // defines one aspect of the appearance of the drawing object	e.g. black line // white fill // black fill //solid (line) // font of letter // colour of triangle	Drawing list	the list of shapes involved in an image // a list that stores the command/description required to draw each object	e.g. triangle // capital letter R // rectangle // line	
Term	Description	Example from logo									
Property	data about the shapes // defines one aspect of the appearance of the drawing object	e.g. black line // white fill // black fill //solid (line) // font of letter // colour of triangle									
Drawing list	the list of shapes involved in an image // a list that stores the command/description required to draw each object	e.g. triangle // capital letter R // rectangle // line									

QUESTION

(b) Riya takes a photograph using a digital camera. The photograph is stored as a bitmap image.

(i) Describe **two** differences between a vector graphic and a bitmap image.

- 1
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-
-
- 2
-
-
-

[4]

MARK SCHEME

5(b)(i)	1 mark for each bullet point to max 2 for each difference • Bitmap made up of pixels // bitmap is made of colours stored for individual pixels • Vector graphic store a set of instructions about how to draw the shape • When bitmap is enlarged the pixels get bigger and it pixelates • When vector is enlarged it is recalculated and does not pixelate • Bitmap files are usually bigger than vector graphics files because of the need to store data about each pixel • Vector graphics have smaller file size because they contain just the instructions to create the shapes • Bitmap images can be compressed with significant reduction in file size • Vector graphic images do not compress well because of little redundant data	4
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Justify the use of a bitmap image or a vector graphic for a given task

9618_w22_qp_12_sc_8.b

QUESTION

(b) State **two** benefits of creating a vector graphic instead of a bitmap image.

1

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2

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[2]

MARK SCHEME

8(b)	1 mark for each bullet point (max 2). For example: • can be enlarged without pixelation / loss of quality • individual components of the image can be edited • generally a smaller file size	2
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Show understanding of how sound is represented and encoded

9618_s25_qp_13_sc_1.b

QUESTION

(b) A teacher records an audio file of the student playing the piano during a music lesson.

Complete the table by giving the term for each description about sound representation on a computer.

Description	Term
the number of times the amplitude is measured per time interval	
the number of bits used to store each amplitude measurement	
the type of sound wave before it is recorded by a computer	

[3]

MARK SCHEME

1(b)	1 mark for each correct term:	3								
	<table><tr><th>Description</th><th>Sound term</th></tr><tr><td>the number of times the amplitude is measured per time interval</td><td>Sampling rate</td></tr><tr><td>the number of bits used to store each amplitude measurement</td><td>Sampling resolution</td></tr><tr><td>the type of sound wave before it is recorded by a computer</td><td>Analogue</td></tr></table>	Description	Sound term	the number of times the amplitude is measured per time interval	Sampling rate	the number of bits used to store each amplitude measurement	Sampling resolution	the type of sound wave before it is recorded by a computer	Analogue	
Description	Sound term									
the number of times the amplitude is measured per time interval	Sampling rate									
the number of bits used to store each amplitude measurement	Sampling resolution									
the type of sound wave before it is recorded by a computer	Analogue									

QUESTION

(c) (i) Explain how an analogue sound wave is converted into digital data.

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..... [2]

MARK SCHEME

6(c)(i)	1 mark for each bullet point (max 2) - Value/magnitude/size of the analogue sound wave is measured a set number of times each second/time / at set intervals - Each sample/reading/measurement is given the binary number and stored in sequence	2
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QUESTION

1 (a) State what is meant by **analogue data**.

..... [1]

MARK SCHEME

1(a)	1 mark for: data values that are continuously changing // variable // any value	1
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QUESTION

(b) Draw **one** line from each term to its most appropriate description.

Term	Description
sampling	the number of samples taken per second
sampling rate	taking measurements at regular intervals and storing the values
sampling resolution	the number of bits used to store each sample

[2]

MARK SCHEME

1(b)	1 mark for 1 correct line 2 marks for 3 correct lines <table><tr><th>Term</th><th>Description</th></tr><tr><td>sampling</td><td>the number of samples taken per second</td></tr><tr><td>sampling rate</td><td>taking measurements at regular intervals and storing the values</td></tr><tr><td>sampling resolution</td><td>the number of bits used to store each sample</td></tr></table>	Term	Description	sampling	the number of samples taken per second	sampling rate	taking measurements at regular intervals and storing the values	sampling resolution	the number of bits used to store each sample	2
Term	Description									
sampling	the number of samples taken per second									
sampling rate	taking measurements at regular intervals and storing the values									
sampling resolution	the number of bits used to store each sample									

QUESTION

(c) The video includes a sound recording.

(i) Describe how sound is represented in a computer.

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..... [3]

MARK SCHEME

3(c)(i)	1 mark each - The amplitude is recorded a set number of times a second - Each (instance of an) amplitude is given a corresponding binary number - The binary number (of each amplitude) is saved in sequence	3
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QUESTION

(b) The audio message is recorded with a sampling rate of 50 kHz and a sampling resolution of 16 bits.

The recording is 20 minutes in length.

Calculate the file size of the recording.

Give your answer in megabytes **and** show your working.

Working

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.....

.....

Answer megabytes

[2]

MARK SCHEME

1(b)	1 mark for answer; 1 mark for working. Working: $\begin{aligned} \text{Size} &= 50\text{KHz} * (20 \times 60) * 16 \text{ bits} \\ &= 50\,000 * 1200 * 16 \text{ bits} // 50\,000 * 1200 * 2 \text{ bytes} \\ &= 960\,000\,000 \text{ bits} \\ &= 120\,000\,000 \text{ bytes} \\ &= 120\,000 \text{ kilobytes} \\ &= 120 \text{ megabytes} \end{aligned}$ Answer = 120 megabytes	2
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9618_w22_qp_11_sc_1.d.i

QUESTION

(d) Sound samples are recorded and saved in a file.

(i) State what is meant by **sampling rate**.

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..... [1]

MARK SCHEME

1(d)(i)	the number of samples taken per unit time / per second	1
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Show understanding of the impact of changing the sampling rate and resolution

9618_w25_qp_13_sc_1.a

QUESTION

- 1 (a) A student creates sound recordings.

Making changes to a sound recording may impact the accuracy of the resulting sound file or the size of the file.

Draw **two** lines from each change to the impacts it has on the sound file.

Change	Impact
increase the duration of the recording	the file size gets bigger
	no change in the file size
	the file size gets smaller
increase the sampling rate	the accuracy of the sound file improves
	no change to the accuracy of the sound file
decrease the sampling resolution	the accuracy of the sound file worsens

[3]

MARK SCHEME

1(a)	1 mark for each correctly connected change box, max 3 marks	3
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QUESTION

(d) The digital camera has a microphone which is used to record the sound for the video.

The user changes the sampling rate that the microphone uses from 44.1 kHz to 88.2 kHz.

Describe how this change in sampling rate will affect the performance of the video doorbell.

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.....

..... [3]

MARK SCHEME

2(d)	1 mark each to max 3: • Data transmission to user's smartphone will take longer • ... because there is more data to transmit • The secondary storage device will fill faster • ... fewer videos will be able to be stored long-term // videos are overwritten more often	3
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QUESTION

(b) Explain the impact of changing the sampling resolution on the accuracy of a sound recording.

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..... [3]

MARK SCHEME

6(b)	1 mark for each bullet point (max 3) Increase sampling resolution ... the number of bits used for each sample is increased ... there will be more values available to represent each sample // more amplitudes can be represented ... each binary amplitude/note in the digital recording is closer to the analogue amplitude/note ... quantisation errors are reduced ... the digital soundwave is closer to the original analogue soundwave Decrease sampling resolution ... the number of bits used for each sample is decreased ... there will be fewer values available to represent each sample // fewer amplitudes can be stored ... each binary amplitude/note in the digital recording is further from the analogue amplitude/note ... quantisation errors are increased ... the digital soundwave is less like the original analogue soundwave	3
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9618_s23_qp_13_sc_3.c.ii

QUESTION

- (ii) A second video is recorded. The sound in the second video needs to be more precise.

Explain the reasons why increasing the sampling rate and the sampling resolution will improve the precision of the second recording.

Sampling rate

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Sampling resolution

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[4]

MARK SCHEME

3(c)(ii)	1 mark each; max 2 for rate and max 2 for resolution Sampling rate - There are smaller 'gaps' in the sound wave // sound is recorded more often - Digital waveform is closer to the analogue waveform - The quantisation errors are smaller Sampling resolution - There are more bits per sample // a wider range of amplitudes can be stored - Each binary amplitude/note (in the digital recording) is closer to the analogue amplitude/note - Digital waveform is closer to the analogue waveform - The quantisation errors are smaller	4
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9618_w22_qp_13_sc_1.a

QUESTION

- 1 A digital audio message needs to be recorded.

- (a) Tick (✓) **one** box in each row to identify the effect of each action on the accuracy of the recording.

Action	Accuracy increases	Accuracy decreases	Accuracy does not change
Change the sampling rate from 40 kHz to 60 kHz.			
Change the duration of the recording from 20 minutes to 40 minutes.			
Change the sampling resolution from 24 bits to 16 bits.			

[2]

MARK SCHEME

1(a)	1 mark for one or two correct ticks, 2 marks for three correct ticks.			2
	Action	Accuracy increases	Accuracy decreases	Accuracy does not change
	Change the sampling rate from 40 kHz to 60 kHz.	✓		
	Change the duration of the recording from 20 minutes to 40 minutes.			✓
	Change the sampling resolution from 24 bits to 16 bits.		✓	

9618_w22_qp_12_sc_6.b.i

QUESTION

(b) The student records a sound file.

(i) Explain the effect of increasing the sampling rate on the accuracy of the sound recording.

.....

 [2]

MARK SCHEME

6(b)(i)	1 mark for each bullet point (max 2): • improves the accuracy of the sound file • ... because (digital) waveform more closely resembles the analogue waveform • quantization errors are reduced • increases the amount of detail stored	2
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QUESTION

- (ii) Explain the effect of decreasing the sampling resolution on the file size of the sound recording.

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..... [2]

MARK SCHEME

6(b)(ii)	1 mark for each bullet point: - decreases the file size of the sound file ... because fewer bits are used to store each sample	2
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QUESTION

- (ii) Explain the effect of increasing the **sampling resolution** on the sound file.

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..... [2]

MARK SCHEME

1(d)(ii)	1 mark for each bullet point (max 2): - increases the number of bits per sample // larger range of values - which means that the file size increases - makes the sound file more accurate //digital waveform closer to original (analogue) waveform - smaller quantisation errors	2
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QUESTION

7 Bobby is recording a sound file for his school project.

(a) He repeats the recording of the sound several times, with a different sample rate each time.

(i) Describe the reasons why the sound is closer to the original when a higher sample rate is used.

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..... [2]

MARK SCHEME

7(a)(i)	1 mark per bullet point - Smaller time gaps between the samples - Makes the digital sound wave more accurate - Smaller quantisation errors	2
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QUESTION

(ii) Describe the reasons why the sound file size increases when a higher sample rate is used.

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..... [2]

MARK SCHEME

7(a)(ii)	1 mark per bullet point - More samples/data are taken/recorded ... so more bits are stored altogether	2
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